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Water loss and shrinkage prediction in 3D printed concrete with varying w/c and specimen sizes

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ABSTRACT

In 3D printed concrete (3DPC), the absence of formwork leads to increased shrinkage compared to traditional cast concrete which uses formwork. This increase in shrinkage is primarily due to the rapid evaporation of moisture during the early stages. Therefore, our research focused on how the initial water content and the exposed surface area affect the total shrinkage by varying water-to-cement ratio (w/c) and the surface-to-volume ratio (s/v). For 3DPC with high w/c, we used calcium formate (CaF) to improve its printability by increasing static yield stress. Dry environment makes the evaporation the predominant mode of water loss, significantly decelerating the hydration process in samples with fully exposed surfaces. At 28 d, the evaporated moisture in 3DPC is 2.8 times higher than the moisture consumed by hydration. A predictive model for the shrinkage properties of 3DPC has been developed by considering that 3DPC experiences both hydration and evaporation simultaneously from early stages on the basis of the water mass conservation principle. The predictions were found to align with actual shrinkage test results, as evidenced by quantitative analysis of parameters like internal RH, mass loss, degree of hydration, and porosity, thus validating the accuracy of the model.

1. Introduction

3D concrete printing (3DCP) has been developing for more than 20 years, accompanied by an increasing amount of researches and applications [1–4]. This innovative fabrication technique brings advantages in terms of safety, economy, efficiency, and design flexibility [5,6]. The 3DCP process can be categorized into three types: Contour Crafting, D-Shape (Monolite) and Concrete Printing [7]. This article solely considers extrusion-based 3D concrete printing, an automated construction technique involving the extrusion and layering of filaments to create formwork-free structures [8,9].

The absence of formwork immediately after deposition results in early surface drying and thus high evaporation rate [10]. The continuous decrease in the menisci radius of pores, caused by evaporation, leads to an increase in capillary pressure [11]. Consequently, the structures fabricated using this innovative method are susceptible to shrinkage and potential cracking when subjected to restrictions [12]. Aggressive chemicals can infiltrate the concrete through these cracks, diminishing mechanical properties and durability and jeopardizing structural security [13]. Therefore, it is crucial to investigate the water loss of 3DPC under drying conditions and its impact on shrinkage behavior. This research is of significance for effectively controlling water loss and minimizing the shrinkage development and cracking risks of concrete structures.

Researchers have extensively investigated the shrinkage properties of 3DPC from various perspectives. Moelich et al. [13] employed digital image correlation (DIC) to study the plastic shrinkage of 3D printed filaments and found that the plastic shrinkage of 3DPC with high surface exposure is several times higher than that of conventional concrete at 25 ± 1 °C and 60 ± 5% RH. Putten et al. [14] applied DIC to explore the unprotected shrinkage of 3DPC and obtained comparable results in both the longitudinal and transverse directions of single filament at 20 ± 3 °C and 65 ± 5% RH. Han et al. [15] utilized two-dimensional DIC to investigate the impact of self-weight from upper layers on the plastic shrinkage of 3DPC. The findings indicated that the horizontal plastic shrinkage of specimens rises with the increase in the self-weight of upper

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layers. From the longstanding shrinkage perspective, Le et al. [16] compared the total shrinkage of the cast 3D printable concrete specimen under different curing conditions. The shrinkage of the specimen at 20 °C, 60% RH is 3–4 times higher than that cured at 20 °C and 100% RH. Slavcheva et al. [17] measured the total shrinkage of cast 3DPC with the dimension of 10 × 40 × 160 mm^3 under varying relative humidity. The results indicated that reducing the RH from 70 % to 20 % resulted in 2.3 times increase in total shrinkage. Table 1 summarizes the key information from these researches.

Various potential mitigation methods have been considered to inhibit the shrinkage of 3DPC. Moelich et al. [18] investigated five shrinkage mitigation methods within 3-h period. They concluded that adding 0.6 vol% short polypropylene fiber (6 mm length) to 3DPC yields the best performance among all the mitigation methods, with 43 % reduction in shrinkage strain. Replacing the Portland cement with 8 % calcium sulfoaluminate cement (CSA) reduces the shrinkage strain by 24 %. The use of superabsorbent polymers (SAP) and shrinkage-reducing admixture decreases the shrinkage strain by 10 % and 6 %, respectively. In terms of curing methods, the application of curing agent reduces the free shrinkage by 34 %. Spraying the water on the specimens before activating the evaporation condition reduces the free shrinkage by 6 %. Putten et al. [14] used superabsorbent polymers in 3DPC to reduce unprotected shrinkage by up to 2 times. Zhang et al. [19,20] incorporated 10–20% recycled brick powder to reduce the drying shrinkage of 3DPC by 3.3 %–12.4% in 60 d.

Although many researchers have investigated the impact of various factors on 3DPC shrinkage, including interlayer generated during layer-by-layer printing, exposure to the environment immediately after printing, and the materials optimization, the effect of high surface exposure on water loss and its relation with shrinkage of 3DPC under drying condition remains unclear. Actually, water loss plays a significant role in the performance of cementitious materials [21]. Water content not only affects the workability of fresh concrete but influences the hydration process, shrinkage and mechanical properties [22]. The typical w/b for 3DPC is around 0.3 [23,24]. However, low w/b requires about 30% more cementitious materials compared to the w/b of 0.5 with the same type of cement, thus being responsible for relatively higher CO_2 emission and cost [25]. On the other hand, specimens with high w/b are more susceptible to the dry environment due to the increased moisture evaporation [26]. Therefore, this research considers the effect of w/c (0.3, 0.4 and 0.5) on the properties of 3DPC.

Another factor affecting the drying shrinkage of 3DPC is the exposure condition [27]. The highly exposed surface of concrete induces more moisture evaporation and higher strain development [28]. Such effects may result in more volume deformation, particularly for 3DPC with highly exposed surface at early stage [29]. Samouh et al. [30] found that the drying shrinkage of cylinders with 0.102 s/v is higher than that with 0.049 s/v. Increased evaporation rates and higher drying shrinkage of the higher s/v concrete can explain this phenomenon.

Although existing research points to an increased tendency for shrinkage in 3DPC as it loses moisture, there is a need for a more comprehensive understanding of the key factors driving this phenomenon. Specifically, further study is required to explore how the initial mixed water content and the surface area exposed to drying conditions influence shrinkage in 3DPC. Therefore, this study examines how w/c and s/v affect the total shrinkage of 3DPC. The yield stress and plastic viscosity were measured to judge its printability. The hydration process of 3D printable concrete with different w/c was monitored by analyzing hydration heat. Long-term properties, including mechanical characteristics, shrinkage behavior, and pore size distribution, were studied over 90 d. The water loss in concrete was characterized through the internal RH, mass loss, and degree of hydration. Additionally, the pore structure was also determined to establish the correlation between capillary pore and shrinkage properties of 3DPC. In this way, the impact of w/c and s/v on the properties of 3DPC was systematically investigated and compared with cast specimens. Additionally, the comparison of the evaporated and hydrated moisture was assessed to understand the effect of early exposure on the water loss of 3DPC. A shrinkage model was developed for 3DPC, taking into account its unique manufacturing characteristics.

2. Total shrinkage modeling of cementitious materials

The modeling for the total shrinkage of 3DPC is intended to be established by drawing on the principles used in the modeling of cast concrete, considering the features of 3DPC. The concrete shrinkage model is formulated based on the mechanisms of shrinkage within different RH environments [31]. Specifically, at RH of 60%, as considered in this research, shrinkage in concrete is primarily driven by capillary pressure [32]. The Biot-Bishop model, which is centered around the concept of effective stress (a principle originally developed in geomechanics) is employed to address shrinkage mainly due to capillary pressure. The notion of effective stress, fundamental to concrete shrinkage models, marks a significant extension of the classical theory of deformable solids to the theory of saturated, deformable porous solids [33]. Effective stress accounts for both the fluid pressure within the material and the externally applied loads [34]. Therefore, in the context of saturated porous materials like concrete, the effective stress (σ'_{ij}) is described by Eq. (1):

$$\sigma'_{ij} = \sigma_{ij} - \zeta p \delta_{ij} \tag{1}$$

$$\zeta = 1 - K_b / K_s$$

where, σ_{ij} is the external stress; ζ is the Biot coefficient, which is related to the bulk modulus; p is the pore pressure; δ_{ij} is the Kronecker delta; K_b is the bulk modulus of the porous solid; and K_s is the bulk modulus of the

Table 1
Key information of shrinkage for 3DPC from various investigations.

Author	Mixture	Method	Specimen dimension	Curing condition	Exposed condition	Main results
Moelich [13]	Mortar with 0.3 w/b	DIC	300 mm in length, 20-layer (printed)	25 ± 1 °C; 60 ± 5%RH with wind speed of 13 ± 1 km/h	Directly exposed to environment after printing	-Propose a crack propagation method-Higher strain and earlier crack of 3DPC
Putten [14]	Mortar with 0.37 w/c	DIC	Single-layered filament, 150 mm in length (printed)	20 ± 3 °C; 65 ± 5%RH	Directly exposed to environment, Measured at 10 min after the addition of water until 7 d	Addition of SAP mitigates the shrinkage of printed concrete up to 200%
Han [15]	Mortar with 0.36 w/b	DIC	250 × 80 × 60 mm^3 (printed)	25 ± 1 °C; 60 ± 2%RH	Directly exposed to environment, measured stopped at 140 min after loading the upper layers	Both free and constrained plastic shrinkage of specimens rises with the increase of upper layers
Le [16]	Mortar with 0.26 w/b	BS EN 12617–4:2002	75 × 75 × 229 mm^3 (cast)	20 °C; 60%RH	Removed mold after one day and monitored over six months	Shrinkage of specimen cured at 60% RH is 3–4 times higher than those cured at 100% RH
Slavcheva [17]	Mortar with 0.4 w/c	Comparator with indicating gage	10 × 40 × 160 mm^3 (cast)	20 ± 2 °C, RH = 20, 30, 50, 70 %	Removed mold after one day and monitored over one month	Total shrinkage increases by 2.8 times changing RH from 70 % to 20 %

solid. In the absence of externally applied stresses, the effective stress for concrete derived from the constitutive equation can be simplified to Eq. (2) [35]:

$$\sigma_v - \left(1 - \frac{K_b}{K_s}\right)p = K_b \varepsilon_v \tag{2}$$

where, σ_v is the volumetric stress; ε_v is the volumetric strain. By reorganizing Eq. (1) into Eq. (2), the volumetric strain equation for a saturated porous concrete can be represented, which is known as the Biot volumetric strain equation [33]:

$$\varepsilon_v = \frac{\sigma_v}{K_b} - p\left(\frac{1}{K_b} - \frac{1}{K_s}\right) \tag{3}$$

The strain development model for partially saturated concrete is enhanced by Bishop [36], who introduces the concepts of liquid and gas phase pressures. This advancement evolves the Biot theory for saturated porous concrete to partially saturated porous concrete. Due to the difficulty in quantifying the effective contact area between the fluid and pore walls, a function related to porosity, denoted as χ, is introduced. This function can be represented as $\chi = \chi\ (S)$, where S denotes the saturation degree. Since the effective contact areas of the liquid and gas phases sum to 1, i.e., $\chi_W + \chi_a = 1$, the effective pore pressure can be described using only the effective liquid phase contact area χ_W, commonly known as the Bishop parameter [37], as shown in Eq. (4):

$$p = \chi_w p_w + (1 - \chi_w) p_a \tag{4}$$

where, p_W is the liquid pressure, GPa; p_a is the gas pressure, GPa. Eq. (4) is integrated into Eq. (3), leading to the formulation of constitutive equation for partially saturated system. This equation, based on the concept of effective stress, is shown in Eq. (5):

$$\varepsilon_v = \frac{\sigma_v}{K_b} - (S_w p_w + (1 - S_w) p_a)\left(\frac{1}{K_b} - \frac{1}{K_s}\right) \tag{5}$$

Capillary pressure is defined as the difference between the gas and liquid phase pressures, i.e., $p_c = p_a - p_W$. It is assumed that the pores of arbitrary shapes are interconnected and uniformly distributed, and in a partially saturated state, the liquid exerts tensile stress within the pore due to capillary menisci formation. Given that in dry conditions, the absolute value of the liquid pressure far exceeds gas pressure, the pressure in the gas is negligible in comparison to capillary pressure. Thus, in the absence of loads applied to 3DPC from the outside and only considering shrinkage performance, Eq. (5) simplifies to Eq. (6), forming the Biot-Bishop model:

$$\varepsilon_v = S_w p_c \left(\frac{1}{K_b} - \frac{1}{K_s}\right) \tag{6}$$

where, P_c is the capillary pressure, GPa; S is the saturation degree; K_b is the bulk modulus of the porous concrete, GPa; K_S is the bulk modulus of the solid.

3. Materials and methods

3.1. Raw materials

The materials used for preparing 3D printable mixtures included ordinary Portland cement P·II 52.5, limestone powder, sand, superplasticizer, calcium formate (CaF), viscosity modifying agent (VMA), and water. P·II 52.5 Portland cement was used as the sole binding material for 3DPC. Calcium limestone powder was used as an inertia material to partially substitute cement, thereby increasing the actual w/c of mixtures. The chemical compositions of cement and limestone powder are listed in Table 2.

The particle size distributions of cement and limestone powder, determined using a laser particle size analyzer, are displayed in Fig. 1. Additionally, the specific surface areas (SSA, determined by the Brunauer-Emmett-Teller (BET) method), as well as the median diameter ($d_{v,50}$) and density of cement and limestone powder, are shown in Table 3.

Silica sand with a maximum particle size of 1.18 mm was adopted as the aggregate. CaF was used to increase the static yield stress of the material with high w/c to attain suitable printability. The purity, melting point, and density of CaF are 98%, 300 °C, and 2.02 g/cm^3, respectively. The VMA with a viscosity of 6000 mPa s, provided by Shandong Head Group Co., Ltd, was employed to increase the green strength and thixotropy of 3D printable concrete.

3.2. Mixture proportions

Table 4 presents the mix proportions of 3DPC with w/c of 0.3, 0.4, 0.5. To satisfy the printability of 3D printable concrete, 0.3% superplasticizer was added by weight of binder for the mixture with 0.3 w/c. CaF (2 % by weight of cement) was added into the mixture of 0.4 and 0.5 w/c groups to increase the static yield stress at early stage and satisfy the printability of the high w/c mixtures. To replace cement, 20% limestone powder was added to the mixture with 0.5 w/c. The s/v ratio remains consistent for all mix compositions.

After dry-mixing the solid materials (cement, limestone powder, sand, PCE, CaF, and VMA) for 1 min, water was added, and the mixture was then wet-mixed for 4 min at the speed of 48 r/min. The specimens were printed and exposed directly under 60% ± 3% RH and 20 °C ± 2 °C environment. After the final setting time (5 h), the specimens were transferred to the constant temperature and humidity chamber (RH = 60% ± 2%, T = 20 °C ± 1 °C).

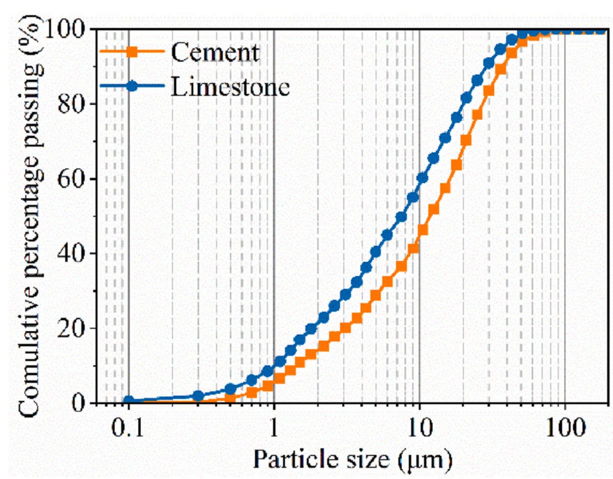


Fig. 1. Particle size distribution of cement and limestone powder.

Table 2
Chemical compositions of cement and limestone powder.

	CaO	SiO_2	Al_2O_3	Fe_2O_3	MgO	SO_3	K_2O	Na_2O	TiO_2	LOI
Cement	64.55	17.47	3.82	2.96	1.00	3.46	0.65	0.26	0.21	2.40
Limestone powder	53.81	1.08	0.43	0.21	0.47	0.02	0.07	0.08	0.03	43.22

Table 3
Median diameter, specific surface area, and density of cement and limestone powder.

	$d_{v,50}$ (μm)	SSA (m^2/g)	Density (g/cm^3)
Cement	11.82	1.76 ± 0.01	2.93
Limestone powder	8.29	1.17 ± 0.06	2.73

3.3. Printing parameter

The gantry-type 3D concrete printer was used to print elements on a 3 m × 3 m × 3 m platform. A circular nozzle with a diameter of 20 mm was employed, and the printing speed was 150 mm/s. The printed filaments had dimensions of 12 mm in height, 25 mm in width, and 2000 mm in length. The deposition duration for one filament was 13 s. The entire structure was fabricated continuously without interruption.

3.4. Specimen size

In this study, the specimens with three sizes are prepared to quantify the influence of s/v on 3DPC, as shown in Table 5. To facilitate clearer result discussion, specimens are labeled with their thickness, and their s/v follows. For instance, the designation T25–0.17 corresponds to a specimen with a thickness of 25 mm and s/v of 0.17 mm^{-1}. The impact of the s/v on 3DPC is studied in mixtures with low w/c (0.3) and high w/c (0.5). Cast specimens were fabricated using the same batch mixture as for printing, and were vibrated for 2 min. To identify the effect of the exposed condition on the printed specimen, cast specimens were demolded 10 min after molding.

3.5. Methods

3.5.1. Rheological properties

Rheological properties were conducted with Brookfield rheometer (vane size: 20 mm diameter and 40 mm height) within a 500 ml container. The static yield stress was defined as the peak of the shear stress curve obtained by rotating the vane for a duration of 3 min at a shear rate of 0.1 s^{-1} [38]. To determine the dynamic yield stress and plastic viscosity, the vane was first rotated linearly from 0 s^{-1} to 50 s^{-1} within 60 s, and then linearly back to 0 s^{-1} in the following 60 s [39]. The dynamic yield stress and plastic viscosity with different w/c were obtained by fitting the downward section of the shear stress-shear rate curve based on Bingham model [40]:

$$\tau = \tau_0 + \mu\gamma \tag{7}$$

where τ is the shear stress, Pa; τ_0 is the dynamic yield stress, Pa; μ is the plastic viscosity, Pa·s; γ is the shear rate, s^{-1}. The static yield stress and dynamic yield stress were tested at 10-min intervals up to 60 min.

3.5.2. Hydration heat

The heat flow and the cumulative heat of cement pastes were monitored for 72 h using the TAM AIR isothermal calorimeter with an accuracy of 20 μW. The admix ampoules with around 10–15 g pastes were placed into an isothermal calorimeter and sealed by the headspace cap for testing [41]. The equipment and the surrounding environment were maintained at 20 °C ± 0.02 °C. The data was collected 10 min after blending with water, which corresponded to the initial measured time of rheological properties.

To continuously monitor the cement hydration process within concrete, the adiabatic temperature is calculated using the data on accumulated heat release [42], as shown in Eq. (8):

$$\alpha = \frac{T(t)}{T(\infty)} \tag{8}$$

where, $T(t)$ is the adiabatic temperature rising at time t, °C;$T(\infty)$ is the adiabatic temperature rise as the cement is completely hydrated, which can be estimated according to Eq. (9):

$$T(t) = \frac{Q}{Cm} \tag{9}$$

where, Q is heat release during the hydration, J/g; C is specific heat capacity of concrete, J/g °C; m is mass of specimen, g.

$$C = 1.05(x_w \bullet C_w + x_c \bullet C_c + x_l \bullet C_l) \tag{10}$$

where, x_w, x_c, x_l are the mass fraction of water, cement and limestone, respectively; C_w, C_c, C_l are the specific heat capacity of water, cement and limestone, respectively.

$$T(\infty) = \frac{m_b}{m_c}\frac{H_0}{C} \tag{11}$$

where, m_b and m_c are the mass of binder and concrete material; H_0 is theoretical heat release when the unit mass cementitious material is completely hydrated.

$$H_0 = 500x_{c_3s} + 260x_{c_2s} + 866x_{c_3A} + 420x_{c_4AF} + 624x_{S0_3} + 1186x_{FreeCaO} + 850x_{MgO} \tag{12}$$

where, x is the mass fraction of different phases.

3.5.3. Mechanical properties

The flexural and compressive strength were tested complying with the Chinese national standard GB/T 17,671–1999 [43]. The dimensions of the specimen for flexural and compressive strength tests were 40 × 40 × 160 mm^3 and 40 × 40 × 40 mm^3, respectively. Every three specimens were tested as a group for flexural strength, and six for compressive strength. The average of each group is calculated as the result. The loading directions for the tests are illustrated in Fig. 2. The printed specimens were extracted from the printed element, which consisted of two parallel filaments and four vertical layers. Both the printed and cast specimens were cured at 20 ± 2 °C and 60% RH, and tested at 3 d, 7 d, and 28 d.

3.5.4. Elastic modulus

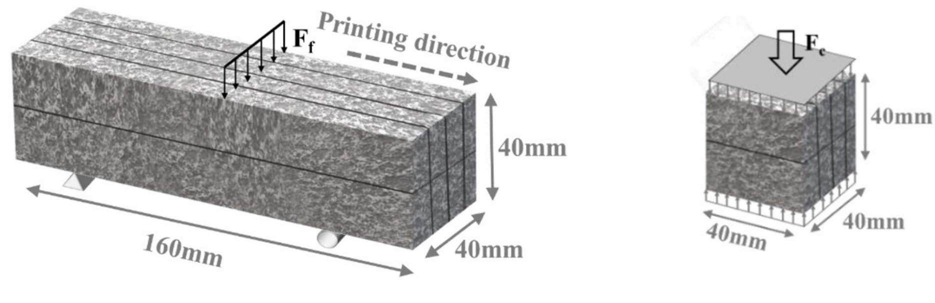
To establish the shrinkage model for 3DPC, the static elastic modulus is selected as the key mechanical parameter of the concrete. The determination of the static elastic modulus followed the Chinese stan-

Table 5
Specimen dimensions.

Notation	Size (mm)	Surface area (mm^2)	Volume (mm^3)	s/v (mm^{-1})
T25–0.17	25 × 25 × 280	29,250	175,000	0.17
T40–0.11	40 × 40 × 160	28,800	256,000	0.11
T70–0.06	70 × 70 × 280	88,200	1,372,000	0.06

Table 4
Mix proportion of 3D printed concrete (kg/m^3).

w/c	Cement	Limestone powder	Sand	Water	CaF	Superplasticizer	VMA
0.3	761.2	0.0	1217.8	228.3	0.0	2.3	0.4
0.4	685.0	0.0	1096.0	274.0	13.7	0.0	0.4
0.5	545.9	136.5	1091.7	272.9	10.9	0.0	0.4



(a) (b)

Fig. 2. Loading of specimens: (a) flexural; (b) compressive test.

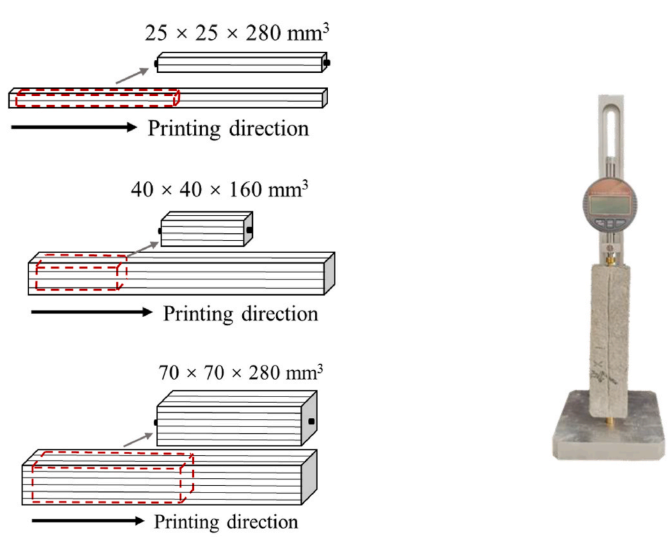
dard JTG E30-2005 [44]. The specimen dimensions were 40 × 40 × 160 mm^3, and tests for the elastic modulus were conducted at 3 d, 7 d, and 28 d. The static elastic modulus can be calculated using Eq. (13):

$$E_s = \frac{F_a - F_0}{A} \times \frac{L}{\varepsilon_a - \varepsilon_0} \tag{13}$$

where, F_a is the load corresponding to $1/3f_{cp}$, N; F_0 is the load corresponding to 0.5 MPa, N; A is the compressive area of the specimen, mm^2; L is the measuring gauge length, mm; ε_a is the deformation of the specimen between gauge lengths at F_a, mm; ε_0 is the deformation of the specimen between gauge lengths at F_0, mm.

3.5.5. Total shrinkage

The vertical length comparator with an accuracy of 0.001 mm was employed to measure the deformation of hardened specimen, as shown in Fig. 3 (b). Three types of prismatic specimens containing interlayers were extracted from the printed element 10 min after printing to explore the effect of s/v on total shrinkage: 25 × 25 × 280 mm^3, 40 × 40 × 160 mm^3, and 70 × 70 × 280 mm^3. Since the height of one filament was 12 mm and the width was 25 mm, shrinkage test specimens with different sizes were taken from printed elements as illustrated in Fig. 3 (a). The measured direction of 3D printed specimen was in parallel with the printing direction. Just after extraction, the copper heads were inserted in the center of the two ends of the specimens. For comparison, the cast specimens with identical sizes were molded. The demolded time is 10 min after molding.



(a) (b)

Fig. 3. Schematic illustration of sample preparation for shrinkage test: (a) 25 × 25 × 280 mm^3, 40 × 40 × 160 mm^3 and 70 × 70 × 280 mm^3; (b) shrinkage measurement.

The initial length of the printed and cast specimens was recorded at 24 h, and every day in the first week, then at 10 d, 14 d, 28 d, 60 d, and 90 d. To evaluate the repeatability of each group, three parallel specimens were tested and the average results were given.

3.5.6. Internal relative humidity

A relative humidity sensor with an accuracy of 1.8% was used to measure the internal RH of specimens. The quartiles were selected to measure the RH gradient within the specimens. The measurements were taken at different depths for each specimen, as shown in Fig. 4. For the specimen with dimensions of 25 × 25 × 280 mm^3, measurements were taken at 5, 10, and 20 mm deep from the surface, respectively. For the 40 × 40 × 160 mm^3 specimens, measurements were taken at 10, 20, and 30 mm. For the 70 × 70 × 280 mm^3 specimens, tests were conducted at 20, 30, and 50 mm in depth, respectively. Two identical measurement results were recorded continually for 28 d with 1-day intervals. The curing temperature was maintained at 20 ± 3 °C and RH of 60 ± 2%. The internal RH of cast specimens was measured at 10, 20, and 30 mm in depth for the specimen with dimensions of 40 × 40 × 160 mm^3 to compare with the printed specimens.

3.5.7. Mass loss

The mass of specimens was weighed using an electronic scale with a precision of 0.01 g. Both printed and cast specimens with varying dimensions (25 mm, 40 mm, 70 mm thickness) under 0.3, 0.4 and 0.5 w/c were measured. The mass measurements were started just after printing, then measured every day during the first week, and subsequently at 10 d, 14 d, 28 d, 60 d, and 90 d under drying conditions (T = 20 ± 1 °C and RH = 60 ± 2%). At least three specimens were tested for each group.

3.5.8. Thermal gravity analysis

The Netzsch TGA testing machine TG209 was employed to obtain the hydration degree of 3DPC. Test samples were extracted from the top, middle, and bottom sections of the specimen with dimension of 40 × 40 × 40 mm^3, as illustrated in Fig. 5 (a). The specimens were cured at 20 ± 1 °C, 60 ± 2% RH, and tested at 1 d, 3 d, 7 d and 28 d. The specimen was immersed in absolute ethanol for each measurement to stop the hydration reaction and oven-dried at 60 °C for 48 h. Around 7 mg sample was ground and heated from 30 °C to 950 °C at the rate of 10 °C/min in a nitrogen atmosphere [45]. Two parallel samples were tested for each measurement. The non-evaporable water content W_B is calculated as given in Eq. (14) [46]:

$$W_B = Ldh + Ldx + 0.41(Ldc - Ldc_a) \tag{14}$$

where Ldh is the loss of chemically bound water from C–S–H (dehydration reaction), which typically occurs within 105–400 °C; Ldx is the dehydroxylation of Portlandite, which occurs within 400–600 °C; Ldc is the decarbonation of $CaCO_3$ within 600–950 °C; Ldc_a is the decarbonation of anhydrous materials, including dry cement and limestone

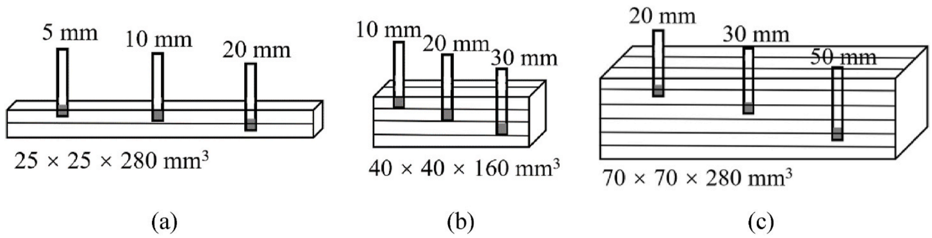


Fig. 4. Internal RH measurement at various depths within three types of specimens.

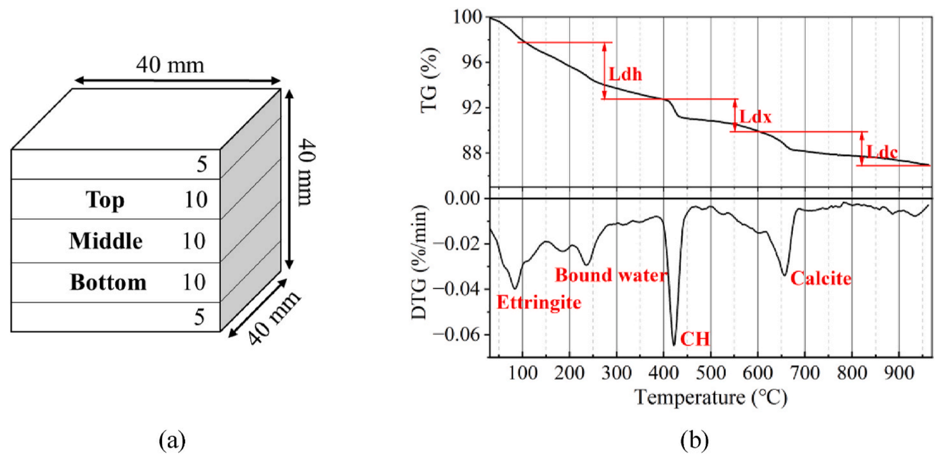


Fig. 5. Thermogravimetric (TG) analysis: (a) sample extracted positions of thermogravimetric analysis; (b) TG-DTG curves of cement paste.

powder. The corresponding typical TG and DTG curves of cement paste are presented in Fig. 5 (b). Once the non-evaporable water content (W_B) is acquired, the degree of hydration α (%) could be calculated using the following equation:

$$\alpha = \frac{W_B}{W_{B\infty}} \times 100 \tag{15}$$

where, $W_{B\infty}$ is the ultimate non-evaporable water content of cementitious material at full hydration, 0.23 [46].

3.5.9. Mercury intrusion porosimetry

The porosity of concrete was analyzed using mercury intrusion porosimetry (MIP). Samples with 3–6 mm diameter were extracted from the center of specimens (40 × 40 × 40 mm^3), which were cured at 20 ± 1 °C and 60 ± 2% RH. Before conducting the MIP test, the samples were removed from the ethanol solution and placed in an oven at a temperature of 55 °C until a constant mass was achieved [47]. The MIP analysis was conducted at 3 d, 7 d, and 28 d. The result was the average of two identical specimens. The minimum detectable pore diameter was 6 nm and the porosity was determined as the ratio of the total pore volume to the specimen volume.

3.5.10. Nitrogen adsorption

Nitrogen adsorption isotherm measurements were conducted to analyze the pore structure with pore diameters below 6 nm. This measurement result helps to capture the entire pore range of the sample under RH of 60%. The measurements were carried out using an Autosorb-IQ2 apparatus of Quantachrome Instrument. Approximately 0.5 g of the sample was cured under 20 ± 1 °C and 60 ± 2% RH. To maintain stability, the samples were placed in a vacuum environment for 24 h before the experiment. The pore diameter available ranged from 1.4 to 170 nm.

4. Results and discussion

4.1. Rheological properties

4.1.1. Static yield stress

The influence of w/c on the initial static yield stress of 3DPC is depicted in Fig. 6 (a). To ensure the printability of the concrete, adjustments to the static yield stress were made for different w/c (0.3, 0.4, and 0.5) to achieve an approximate value of 2000 Pa. Specifically, 0.3% addition of superplasticizer to the cement content was utilized for the material with w/c of 0.3.2% CaF of the cement content was incorporated for materials with w/c of 0.4 and 0.5. To further increase the w/c, the cement was replaced by 20% limestone powder for the composition of 0.5 w/c.

Fig. 6 (b) illustrates the development of static yield stress over time for 3D printable concrete with varying w/c. According to Ref. [48], the growth rate of static yield stress, denoted as A_{thix}, is calculated using the following expression:

$$\tau_0(t) = \tau_0 + A_{thix} t \tag{16}$$

where $\tau_0(t)$ is the static yield stress at time t, Pa; τ_0 is the initial static yield stress (10 min after adding water), Pa.

The experimental data, as demonstrated in Fig. 6 (b), revealed a linear increase in the static yield stress of 3D printable concrete within 1 h. Notably, the static yield stress growth rates for concrete with w/c of 0.3 and 0.4 exhibited similar values at 104.9 Pa/min and 107.4 Pa/min, respectively. These rates were higher than the 80.7 Pa/min observed for concrete with 0.5 w/c. This difference is ascribed to the relatively lower specific surface area of limestone powder (1.168 ± 0.059 m^2/g), in contrast to that of cement (1.762 ± 0.011 m^2/g), as detailed in Table 3. The substitution of 20% of the cement with limestone powder resulted in a reduced specific surface area, diminishing the water demand and resulting in a slower static yield stress growth rate. These findings align with Xiang et al. [49], who indicated that higher specific surface area of

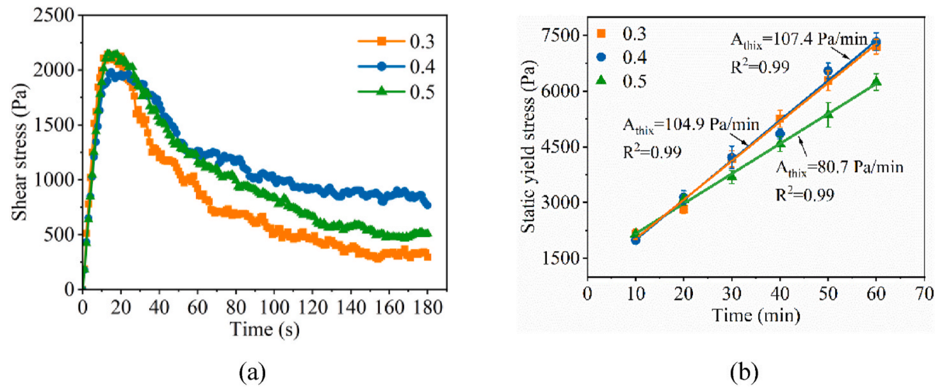


Fig. 6. Static yield stress: (a) shear stress curves with various w/c measured at 10 min; (b) static yield stress evolution from 10 min to 60 min after adding water.

the raw materials correlates with an increased static yield stress. Furthermore, as limestone powder is an inert mineral admixture, the 20% mass replacement of cement has a dilutive effect on cement content, thereby decelerating the hydration process and subsequently leading to a diminished growth rate of static yield stress [50].

4.1.2. Dynamic yield stress and plastic viscosity

As depicted in Fig. 7 (a), the dynamic yield stress and plastic viscosity of 3D printable concrete vary with different w/c. The Bingham model, represented by Eq. (7), showed a strong linear relationship between shear stress and shear rate in the descending shear speed range, with $R^2 > 0.952$. The intercept and slope of this range correspond to the dynamic yield stress and plastic viscosity, respectively. Fig. 7 (b) illustrates the influence of w/c on these properties, indicating a decrease in both dynamic yield stress and plastic viscosity as w/c increases. This trend, particularly for plastic viscosity, can be attributed to a decrease in the effective volume fraction of solid materials within the concrete as the w/c rises [51].

4.2. Hydration heat

In the preceding section, an increase in static yield stress was observed for 3D printable concrete with high w/c after the addition of CaF. To investigate this behavior, we monitored the hydration heat of cement paste over 72 h. The results, depicted in Fig. 8 (a), included assessments of both the mix proportions outlined in Table 4 and variations without CaF. This approach allowed for an isolated evaluation of the effects of w/c and CaF on the hydration heat. From the heat flow curves, two characteristic exothermic peaks during hydration are identified: an initial peak resulting from the cement reaction with water and the dissolution of raw materials and the second peak associated with the formation of C–S–H [52]. The period between these peaks, known as the induction period, was marked by the specific point on the first derivative of the heat flow curve [53]. The main parameters derived from the heat flow analysis are summarized in Table 6.

Analysis of Table 6 reveals distinct differences in the hydration behavior of cement pastes with varying w/c. Pastes with high w/c of 0.4 and 0.5 exhibited earlier ends of the induction period and earlier occurrences of the second exothermic peak compared to those with low w/c of 0.3, indicating a faster hydration process. The addition of CaF to these high w/c pastes further expedited this process. This accelerated hydration is due to the larger available space for hydration products to form, given the increased water content, as reflected by the heightened cumulative heat release over 72 h in the paste with w/c of 0.4, as shown in Fig. 8 (b). Conversely, the paste with w/c of 0.5 showed the least cumulative heat release. This can be attributed to the substitution of 20% of the cement with limestone powder in the 0.5 w/c mixture, diluting the cementitious material and resulting in a lower heat release [50].

The static yield stress must meet the required static yield stress for printability of 3D printable concrete [54]. This study explored the effect of CaF on the rheological properties of 3D printable concrete by examining the hydration heat release rate within the first hour, as shown in Fig. 8 (c). The incorporation of CaF in cement paste resulted in a 20.7% increase in heat flow at the first exothermic peak for the mixture of 0.4 w/c, and a significant 111.1% increase for mixture of 0.5 w/c, compared to mixes without CaF. As depicted in Fig. 8 (d), the mix with w/c of 0.3 without CaF exhibited the highest 1-h cumulative heat release, with 0.4 and 0.5 following behind. The heightened release observed in the 0.3 w/c mixture can be attributed to the higher content of binding materials. However, with the addition of CaF, there was a notable rise in the 1-h cumulative heat release for pastes with high w/c. The data from the hydration heat tests indicate that CaF accelerates the hydration process, thereby increasing the static yield stress of materials suitable for high

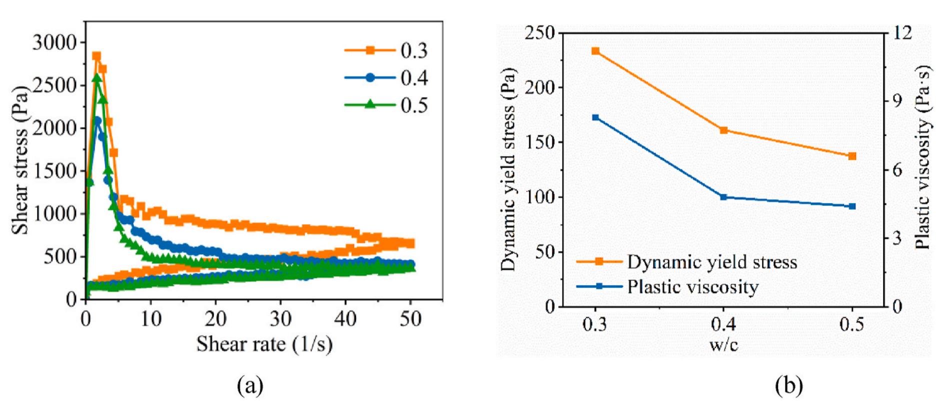


Fig. 7. (a) Effect of w/c on dynamic yielding behaviors; (b) dynamic yield stress and plastic viscosity under different w/c.

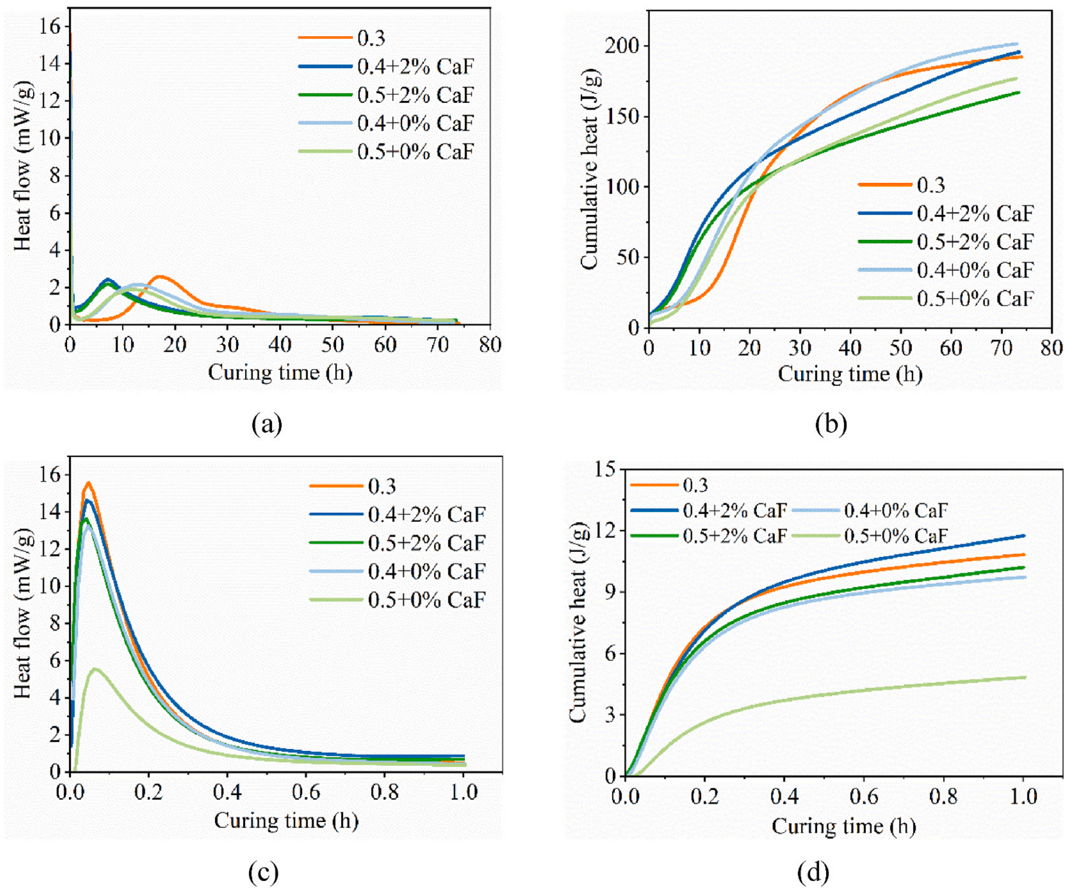


Fig. 8. Hydration heat of cement pastes with various w/c and with/without the additions of CaF: (a) heat flow within 3 d; (b) cumulative heat within 3 d; (c) heat flow within 1 h; and (d) cumulative heat within 1 h.

Table 6
Major parameters obtained from the calorimetry test.

Mixture	Heat flow of the first peak (mW/g)	End time of induction period (h)	Time of second peak (h)	Heat flow of second peak (mW/g)	Cumulative heat at 1 h (J/g)	Cumulative heat at 72 h (J/g)
0.3	15.58	4.78	17.06	2.59	10.84	191.66
0.4 + 2% CaF	14.61	0.87	7.17	2.42	11.75	194.47
0.4 + 0% CaF	13.23	2.04	12.86	2.16	9.74	201.09
0.5 + 2% CaF	13.62	0.84	7.21	2.18	10.21	165.83
0.5 + 0% CaF	5.52	1.86	12.90	1.89	4.84	176.24

Note: 0.3, 0.4, and 0.5 represent the w/c, 0% CaF and 2% CaF represent the addition of CaF content.

w/c 3D printable concrete, with CaF initiating the reaction immediately upon water contact [55].

The observed increase in hydration heat for pastes with high w/c can be attributed to the role of CaF. Upon addition, CaF increases the concentration of Ca^{2+} in the liquid phase, which in turn accelerates the dissolution of C_3A and C_3S in cement [55,56]. Furthermore, the interaction of the formate ion ($HCOO^-$) with C_3A leads to the formation of an AFt phase analog of formate ion, specifically $C_3A{\cdot}Ca(HCOO)_2{\cdot}nH_2O$, which contributes to the increase of the green strength of the paste with high w/c [57]. Consequently, the incorporation of CaF is identified as an effective strategy for increasing the static yield stress, thereby optimizing the printability of 3D printable concrete with high w/c.

4.3. Mechanical properties

The flexural and compressive strengths of printed and cast specimens at 3, 7, and 28 d are shown in Fig. 9. At 28 d, the flexural strength of printed specimens was higher than that of cast specimens, which is related to the loading direction being perpendicular to the printing direction. In flexural tests, the maximum tensile stress is located in the middle of the bottom of the specimen [16]. When loading the flexural strength perpendicular to the printing direction, there is no weak interlayer area at the central bottom position of the printed specimen, so the flexural strength of the printed specimen is higher. The relationship between the loading direction and the printing direction is shown in Fig. 2 (a). As can be known from Fig. 9 (b), the compressive strength of printed specimens was lower compared to cast specimens, mainly because the interlayer interfaces with larger pores that occurred during the layer stacking process of 3D concrete printing weaken the compressive strength of the printed specimens [58].

4.4. Total shrinkage

Fig. 10 (a) shows the effect of the s/v on the shrinkage properties of printed specimens. The total shrinkage increased sharply within the first 10 d and tended to stabilize around 28 d. As the s/v of the specimens

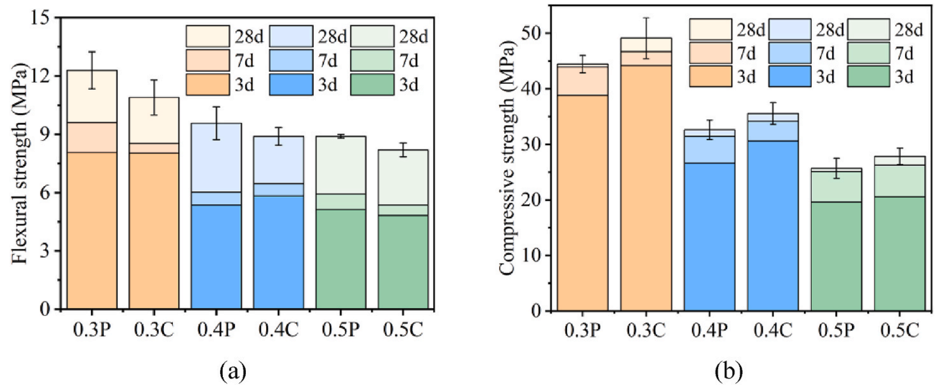


Fig. 9. Effect of w/c on mechanical properties: (a) flexural; (b) compressive strength. (0.3, 0.4, 0.5 refer to the w/c; P: printed specimen; C: cast specimen).

increased from 0.06 (with a thickness of 70 mm) to 0.17 (with a thickness of 25 mm), the total shrinkage of printed specimens with w/c of 0.3 increased by 19.8% at 90 d, and the total shrinkage of printed specimens with w/c of 0.5 increased by 17.5%.

Fig. 10 (b) presents the total shrinkage of printed and cast specimens with varying w/c, the dimension of the specimens is 40 × 40 × 160 mm^3. The results show that an increase in w/c from 0.3 to 0.5 resulted in a maximum increase of 37.3% in total shrinkage for printed specimens at 90 d. This is mainly due to the fact that the higher the w/c, the faster the evaporation of water in the concrete, thus leading to increased drying shrinkage [59]. Additionally, the hydration process contributes to the refinement of pores in specimens with low w/c, further slowing down the migration and the loss of water [26]. Therefore, 3D printed concrete with low w/c exhibits lower shrinkage compared to high w/c specimens. Moreover, the replacement of 20% limestone powder in specimens with w/c of 0.5 resulted in a lower total shrinkage compared to those with w/c of 0.4.

When comparing the differences between printed and cast specimens under the influence of different w/c, it was found that the total shrinkage at 28 d for printed specimens with w/c of 0.3 was similar to that of the cast specimens, with a maximum difference of 4.6%. As the w/c increased from 0.3 to 0.5, the difference in total shrinkage at 28 d between the printed and cast specimens gradually increased to 10.1%. This phenomenon can be explained by the capillary pore volume fraction difference between printed and cast specimens, as detailed in section 4.8.

4.5. Internal RH

The effect of s/v on internal RH in printed specimens with w/c of 0.3 is shown in Fig. 11. Initially, the internal RH of the specimens decreased sharply and then leveled off. We observed a minimal variation in RH levels, no more than 2.4%, across different depths in specimens with low w/c. This uniformity is due to the rapid hydration and evaporation processes of 3DPC with all surfaces exposed. Moisture quickly reaches an equilibrium with the ambient environment, minimizing the RH gradient inside the specimen.

Fig. 12 shows how s/v affects internal RH in 3D printed specimens with a w/c of 0.5. In thinner specimen (25 mm thick), the internal RH was consistent at all depths, with an ignorable 1.9% difference. This implies that internal RH in thinner specimens is almost uniform. However, in thicker specimens (40 and 70 mm thick), the internal RH varies: it is lower at the top and bottom than in the middle. This variation suggests the presence of RH gradient in specimens with higher thickness.

It is important to note that moisture evaporation and hydration are the primary factors affecting water loss in concrete [27]. In specimens with high w/c, there is a lot of water, which tends to move to the surface, creating humidity gradient within the specimen. On the other hand, in specimens with low w/c, more water is consumed in the hydration process, leading to a more even spread of internal RH [60]. This interaction between evaporation and hydration is vital to understanding water loss in concrete. To better understand how each process, evaporation and hydration, affects the water content in 3DPC, we need to calculate the water used in each process separately. We will discuss and compare these calculations in the discussion section 4.1.

Fig. 13 Shows the impact of s/v on the averaged internal RH from measurements taken at the top, middle, and bottom locations. The shaded area in the figure represents the standard deviation. The results indicated that with an increase in the s/v, the internal RH decreased more rapidly in the early stages. Specimens with both w/c of 0.3 and 0.5 exhibited the same trend. Specimens with thicknesses of 25 mm and 40 mm had an internal RH that dropped to about 60% at 28 d, while specimens with a thickness of 70 mm had an internal relative humidity of about 75% at the same age. This suggests that specimens with a

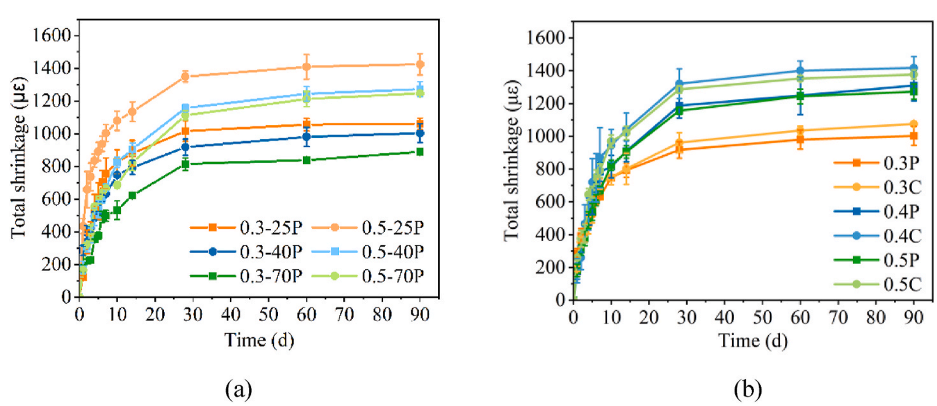


Fig. 10. Total shrinkage of specimens: (a) effect of specimen dimensions (25, 40, 70 represent the thickness of specimens); (b) effect of w/c on the total shrinkage.

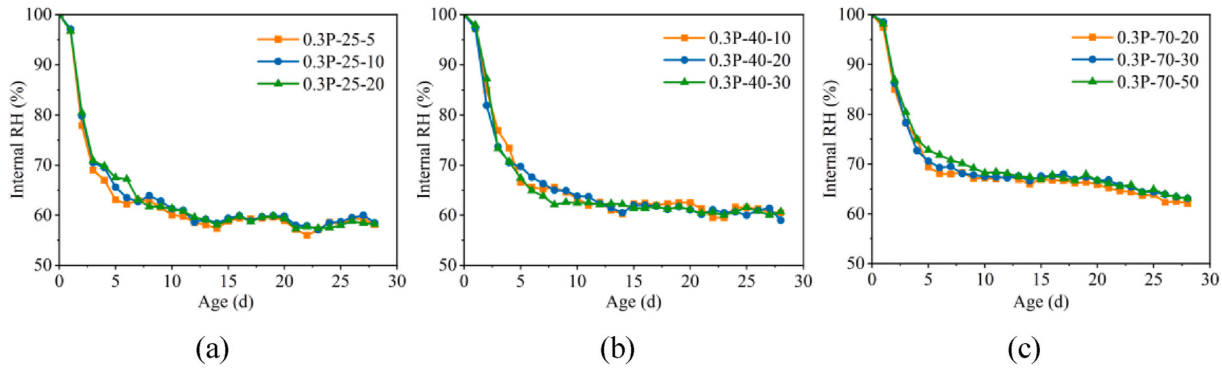


Fig. 11. Evolution of internal RH of the specimens with various s/v measured at different depths under the w/c of 0.3: (a) 25 mm thickness of specimen; (b) 40 mm thickness of specimen; (c) 70 mm thickness of specimen. (The number at the end of the legend represents the measured depth).

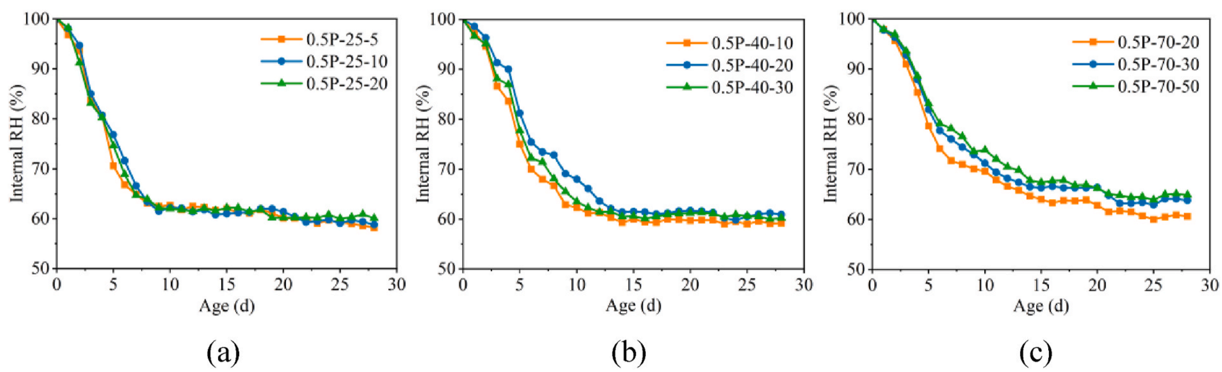


Fig. 12. Evolution of internal RH of the specimens with various s/v measured at different depths under the w/c of 0.5: (a) 25 mm thickness of specimen; (b) 40 mm thickness of specimen; (c) 70 mm thickness of specimen. (The number at the end of the legend represents the measured depth).

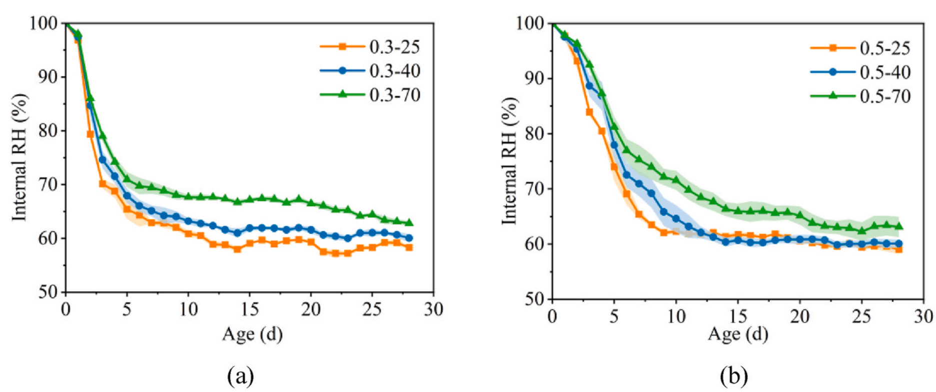


Fig. 13. Effect of s/v on internal RH: (a) w/c = 0.3; (b) w/c = 0.5.

smaller s/v (thickness of 70 mm) have a stronger water retention capacity.

Fig. 14 depicts the influence of the w/c on the internal RH of both printed and cast specimens, each with dimensions of 40 × 40 × 160 mm^3. On the first day, the internal RH for all specimens was around 97%. For all types of specimens, there was a rapid decrease in internal RH within the first 7 d, followed by a stabilizing stage. Notably, specimens with w/c of 0.3 exhibited the quickest decline in RH, with those at 0.4 and 0.5 following. This trend is primarily ascribed to lower initial water content in specimens with low w/c, resulting from the combined effects of evaporation and hydration, which leads to a rapid loss of moisture and a decrease in RH [61]. Concerning the impact of fabrication method, the internal RH of printed specimens across different w/c was closely aligned with that of cast specimens , the difference between two fabrication methods was around 3%.

4.6. Mass loss

Fig. 15 shows the mass loss per cubic meter over time for specimens with varying s/v. As the s/v of the specimen increases, the mass loss per cubic meter of the specimen increases. Furthermore, an increase in the w/c from 0.3 to 0.5 also resulted in increased moisture loss. These observations demonstrate that both the dimension of the specimen (s/v) and its initial water content (w/c) significantly influence the mass loss of specimen in dry environment.

Fig. 16 (a) displays the mass loss in both printed and cast specimens with varying w/c. With an increase in the w/c from 0.3 to 0.5, there was a noticeable increase in water evaporation. In terms of the impact of the

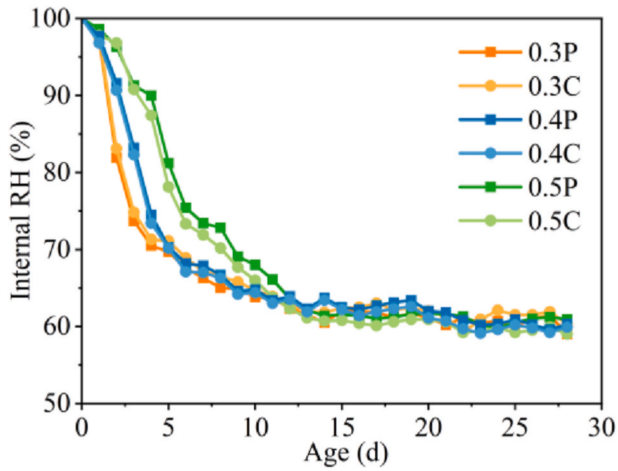


Fig. 14. Effect of w/c on internal RH with various fabrication methods.

fabrication method, the moisture loss in printed and cast specimens was found to be similar, with a maximum difference of 8.9%. Fig. 16 (b) explores the relationship between mass loss per cubic meter in specimens and their total shrinkage. Here, a linear correlation was found between the total shrinkage in 3DPC and mass loss. Continuous moisture loss accelerates the development of capillary pressure, which in turn increases shrinkage strain [62]. Additionally, more significant shrinkage has been observed in the early stage in low w/c specimens. This is primarily because the mass loss in these specimens only accounts for evaporative water loss, not considering water consumed in the hydration process [61]. However, in low w/c concrete, the early-stage water experiences significant water loss due to hydration, resulting in higher shrinkage under the same mass loss conditions.

4.7. Hydration degree

Fig. 17 presents the thermogravimetric (TG) curves of 3DPC with varying w/c at 1 and 28 d. The total weight loss of these samples increased with the increase of the w/c. In cementitious materials, weight loss typically occurs at 400 °C and 700 °C. The weight losses in these two temperature ranges are mainly associated with the decomposition of $Ca(OH)_2$ and $CaCO_3$ [45]. Notably, the samples with w/c of 0.5 demonstrated more pronounced weight loss at 700 °C. This is indicative of increased decomposition of carbonate products during heating, a phenomenon linked to the inclusion of 20% limestone powder in the materials with w/c of 0.5 [63].

The TG methods are conducted on the top, middle, and bottom sections of the specimens, as shown in Fig. 5 (a). At 1 d, the sample from the middle position exhibited the most significant weight loss, followed by the top and bottom positions, which displayed similar results across all w/c. By 28 d, however, the weight loss across these three positions converged. Initially, the top and bottom surfaces of the sample, being exposed to the environment, lost more water and underwent a lower degree of hydration. In contrast, the middle sample retained more moisture internally, leading to a higher degree of hydration. As time progressed, the internal RH across different positions almost equalized, reducing the variance in thermal weight loss within the specimen. The early gradient in the degree of hydration, which disappears later, is consistent with the trend in the internal relative humidity gradient observed in section 4.5.

The hydration degree of 3DPC with different w/c is shown in Table 7. The results are averages from three locations on the specimens (top, middle, and bottom). In the temperature ranges of 105–400 °C, 400–600 °C, and 600–950 °C, the weight loss of samples notably

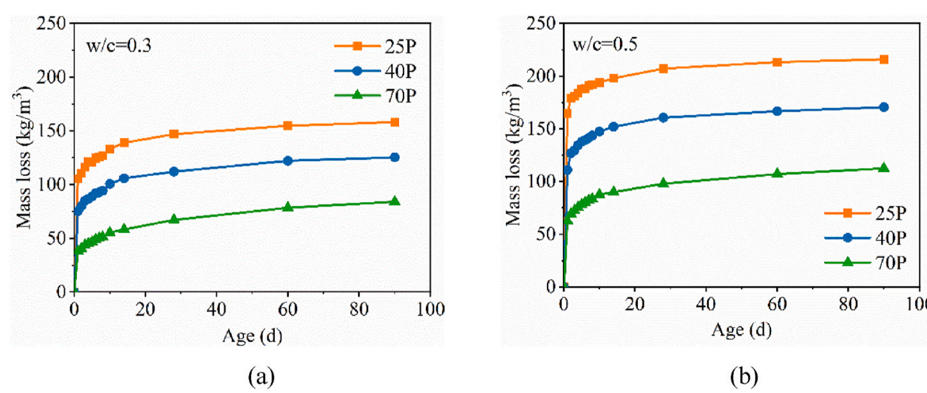


Fig. 15. Mass loss per cubic meter with age: (a) 0.3 w/c; (b) 0.5 w/c.

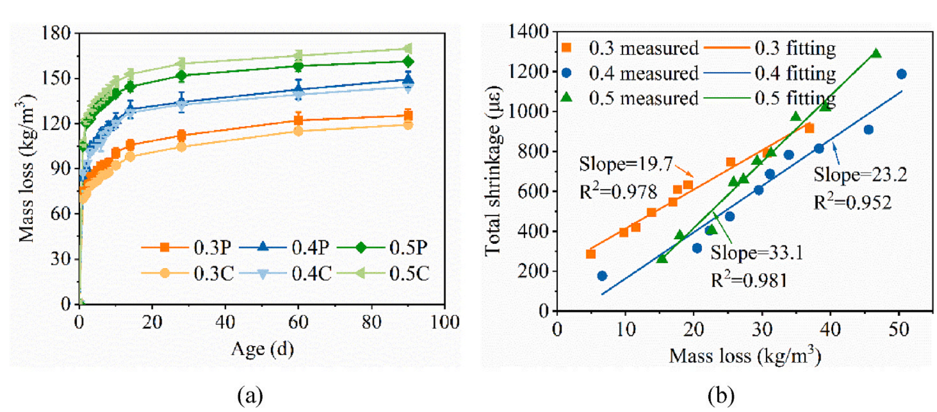


Fig. 16. (a) Effect of w/c on mass loss; (b) correlation between mass loss and total shrinkage.

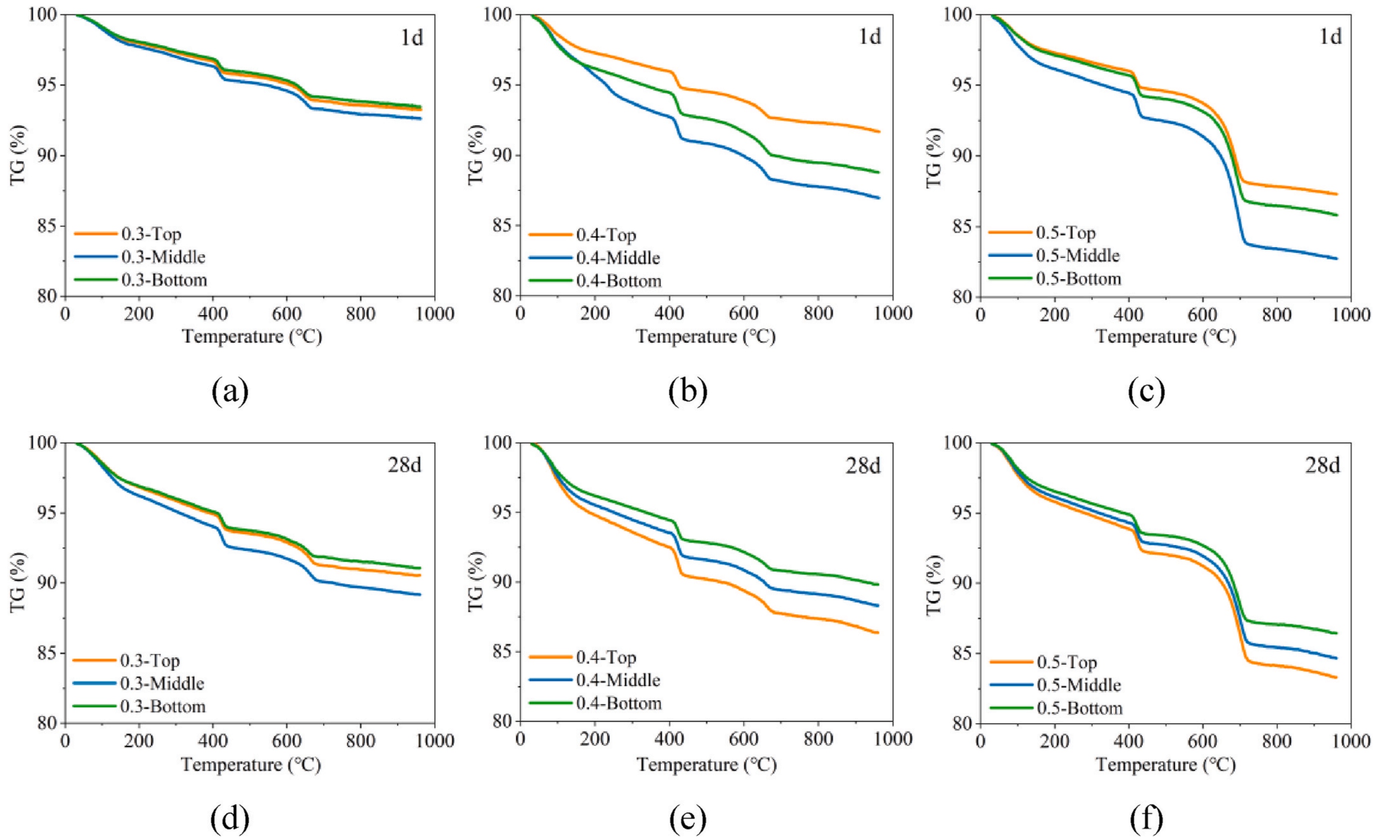


Fig. 17. Thermogravimetric (TG) analysis of 3DPC with various w/c at 1 d: (a) 0.3; (b) 0.4; (c) 0.5 and 28 d: (d) 0.3; (e) 0.4; (f) 0.5.

increased over time, corresponding with increases in Ldh, Ldx, and Ldc (definitions provided in section 3.5.8). This indicates that more C–S–H, Ca $(OH)_2$, and $CaCO_3$ are produced within 28 d, increasing the degree of hydration of the 3DPC. Furthermore, at 28 d, an increase in w/c from 0.3 to 0.5 resulted in a higher hydration degree. This increased hydration degree is due to the higher available water for hydration products to form for high w/c specimens.

4.8. Pore size distribution

The effect of different w/c on the porosity of specimens at 28 d obtained by MIP is shown in Fig. 18 (a). An increase in w/c from 0.3 to 0.5 led to a rise in porosity, which aligns with the mechanical property findings discussed in section 4.3. W/c. Fig. 18 (b) further illustrates the inverse relationship between compressive strength and porosity. Notably, the slope representing porosity against compressive strength was 3.1 for cast specimens and 6.1 for printed specimens. This higher slope observed in printed specimens suggests a more pronounced decrease in compressive strength within the same porosity range, indicating that printed specimens are more susceptible to changes in porosity compared to cast ones. This heightened sensitivity in printed specimens can be attributed to their higher pore connectivity [64].

Table 7
Hydration degree of various w/c at 1 d and 28 d (%).

Age	w/c	Position	Ldh	Ldx	Ldc	Ldca	α (%)	Average α (%)
1 d	0.3	Top	2.28	1.59	1.82	0.02	20.0	20.6
		Middle	2.56	1.74	1.96	0.02	22.1	
		Bottom	2.21	1.53	1.85	0.02	19.5	
	0.4	Top	2.51	2.08	2.19	0.02	23.8	30.2
		Middle	4.20	2.78	2.98	0.02	35.6	
		Bottom	3.24	2.77	2.88	0.02	31.2	
	0.5	Top	2.44	2.27	6.43	0.44	31.2	36.1
		Middle	3.28	3.05	8.65	0.44	42.1	
		Bottom	2.68	2.55	7.32	0.44	35.0	
28 d	0.3	Top	3.35	1.88	2.66	0.02	27.5	30.6
		Middle	3.97	2.17	2.99	0.02	32.0	
		Bottom	3.99	2.24	3.03	0.02	32.5	
	0.4	Top	4.20	2.91	3.38	0.02	36.9	38.3
		Middle	4.42	3.10	3.50	0.02	38.9	
		Bottom	4.46	3.07	3.60	0.02	39.1	
	0.5	Top	3.99	3.06	8.37	0.44	44.8	44.2
		Middle	4.04	2.91	8.54	0.44	44.6	
		Bottom	3.77	2.86	8.42	0.44	43.1	

The minimum tested pore diameter for MIP is 6 nm. Based on the Kelvin equation, the critical pore radius r_c under the relative humidity (RH) environment of 60% in this study can be determined:

$$r_c(\mathrm{RH}) = \frac{-2 \bullet \gamma \bullet V_m}{\ln\left(\frac{\mathrm{RH}}{100}\right) \bullet R \bullet T} \tag{17}$$

where γis the surface tension of water, 0.073 N/m; V_m is the molar volume of water, 1.8×10^{-5} m^3/mol; R is the universal gas constant, 8.314 J/(mol·K); and T is the absolute temperature, 293.15 K.

At a relative humidity of 60%, the critical pore radius r_c can be calculated as 2.08 nm. Therefore, the minimum pore diameter affecting shrinkage is around 4.2 nm. The pore volume between 4.2 and 6 nm is not measurable by MIP. To more clearly understand the impact of pores with diameters of 4.2–50 nm on the shrinkage properties of the specimens, the pore volume was tested using the N_2 adsorption method for a more precise measurement of these smaller pores, as shown in Fig. 19 (a).

Using the N_2 adsorption method allows for the calculation of the capillary pore volume ranging from 4.2 to 50 nm, as detailed in Table 8. The pore volume ranging from 4.2 to 50 nm increases with the increase in the w/c from 0.3 to 0.4. Furthermore, the pore volume from 4.2 to 50

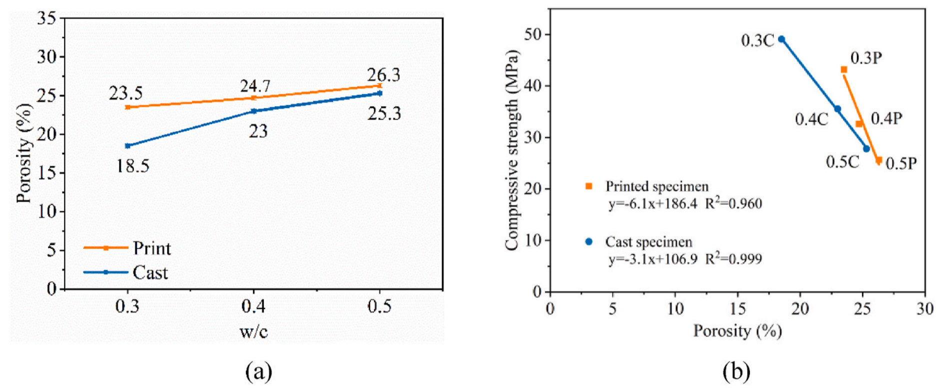


Fig. 18. Porosity of concretes: (a) effect of w/c; (b) its correlation with compressive strength.

nm for the w/c of 0.5 is lower compared to that of 0.4, which is attributed to the filling effect resulting from the replacement of 20% limestone. Printed specimens had a lower capillary pore volume in the 4.2–50 nm range compared to cast specimens. Notably, as the w/c increased from 0.3 to 0.5, the difference in capillary pore volume between printed and cast specimens became more pronounced. This increase in the difference in capillary pore volume suggests that the influence of the fabrication method on the capillary pores becomes more significant as the w/c increases. In printed specimens, compaction occurs instantly during extrusion, while cast specimens undergo a 2-min vibration process. Despite the pressure from upper layers in printed structures, the rapid increase in static yield stress resists this pressure effectively. Consequently, vibration compaction in casting has a more profound effect on pore structure than the extrusion process in 3D printing. This leads to a larger volume of capillary pores compared to printed specimens. Additionally, higher w/c specimens (0.4 and 0.5), with their initially larger pore volumes, are more responsive to the densifying effects of the vibration process of cast specimens, resulting in a more significant initial pore refinement than in printed specimens. This leads to an increased difference in capillary pore volume between printed and cast specimens with rising w/c. Fig. 19 (b) further establishes the relationship between the pore volume in the range of 2.4–50 nm and shrinkage, revealing a good linear correlation between the capillary pore volume in this range and shrinkage.

5. Modeling and analysis

3DPC is often exposed to the environment directly during printing, leading to easy migration of moisture from the concrete to its surroundings. Thus, compared to traditional cast concrete, the early exposure of 3DPC accelerates its moisture evaporation within the concrete [65]. The main mechanisms of water loss in 3DPC are evaporation and hydration [27]. In this study, we calculated the amounts of water lost due to evaporation and hydration respectively. These calculations allow for a comparative analysis of both evaporated and hydrated water in 3DPC.

Table 8
Pore volume (mL/g) ranging from 4.2 to 50 nm of samples.

w/c	Printed sample	Cast sample
0.3	0.011	0.012
0.4	0.017	0.023
0.5	0.014	0.018

5.1. Comparison between the evaporated and hydrated water of 3DPC with varying w/c

Water loss in concrete is comprised of two components: hydration and evaporation. Hydration-related water loss was calculated through the degree of hydration, while evaporation-induced water loss was determined by measuring mass loss. The specific calculation of the hydrated water amount W_{hyd} is given by Eq. (18):

$$W_{hyd} = \alpha(t) \bullet W_{\infty} \bullet m_{ce} \tag{18}$$

where $\alpha(t)$ is the averaged degree of hydration of cementitious material, determined from measurements collected at the top, middle, and bottom locations of the specimens, obtained by thermogravimetric analysis, as detailed in section 4.7; W_{∞} is the amount of water required for complete hydration of concrete per unit mass of concrete, 0.23 kg/kg; m_{ce} is the mass of cement per unit volume of concrete, kg/m^3, see Table 4. The evaporated water content W_{eva} can be expressed by Eq. (19):

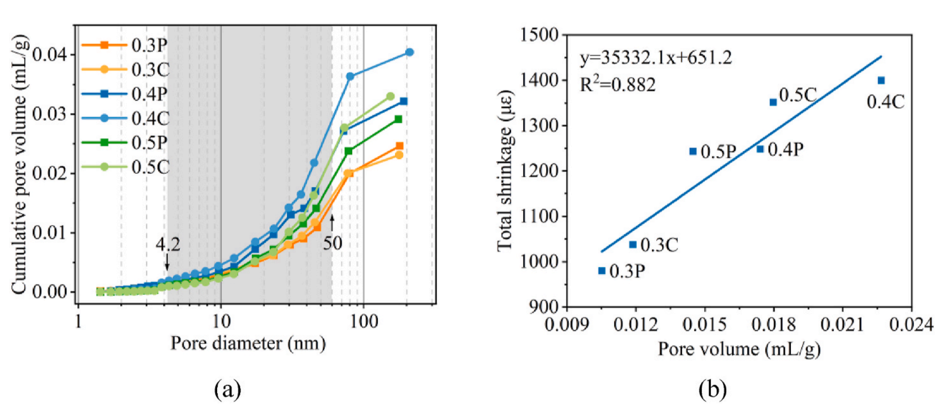


Fig. 19. Pore structure measured by BET test: (a) pore volume fraction for printed and cast specimens with various w/c; (b) its correlation with total shrinkage.

$$W_{eva} = \frac{\Delta m}{V_{sample}} \tag{19}$$

where Δm is the mass loss of water per unit volume of concrete, kg/m^3; V_{sample} is the volume of the specimen, m^3.

The separate calculations for water loss due to hydration and evaporation in 3DPC are carried out using Eq. (18) and Eq. (19). As illustrated in Fig. 20, it is evident that the water loss attributable to evaporation consistently exceeds that due to hydration. Specifically, at 28 d, the water loss through evaporation was found to be up to 2.8 times higher than that lost through hydration. This trend highlights evaporation as the dominant factor in the water loss of 3DPC. Additionally, our findings indicate that the proportion of water loss attributed to evaporation increased with the increase in w/c, rising from 68% to 74% at 28 d. This suggests that the relative water loss proportion due to hydration decreases with the increase in w/c.

During the initial 3 d, water loss in 3DPC significantly increased due to both evaporation and hydration. In contrast, traditional cast concrete, typically not demolded until at least one day of curing, early water loss primarily resulted from hydration. This difference substantially reduces the initial water loss in concrete. Wang [66] observed that specimens exposed to dry environment (25 ± 2 °C and 30% RH). Consumes twice as much water as those exposed after 3 d, highlighting the protective effect of formwork against early evaporation in traditional cast concrete. In comparison, 3D printed specimens, being exposed to the environment sooner and over larger surface areas, exhibited reduced hydration, leading to coarser pores and potentially decreased mechanical properties and durability. Therefore, for 3DPC, implementing measures such as increasing the RH of environment during printing and covering the surface for curing is recommended to reduce rapid early water evaporation.

5.2. Correlation between water loss and total shrinkage

Fig. 21 illustrates how overall water loss, including evaporated and hydrated water, relates to the total shrinkage for 3DPC specimens with w/c of 0.3, 0.4, and 0.5. Our findings indicate that the total shrinkage increases linearly with the amount of water consumed. Notably, specimens with 0.3 w/c did not exhibit a significant increase in total shrinkage during the early stage compared to those with 0.4 and 0.5 w/c. This observation contrasts with the data presented in Fig. 16 (b), which solely accounts for water loss due to evaporation. This discrepancy suggests that incorporating both evaporated and hydrated water in the analysis provides a more accurate relationship between water loss and total shrinkage in 3DPC.

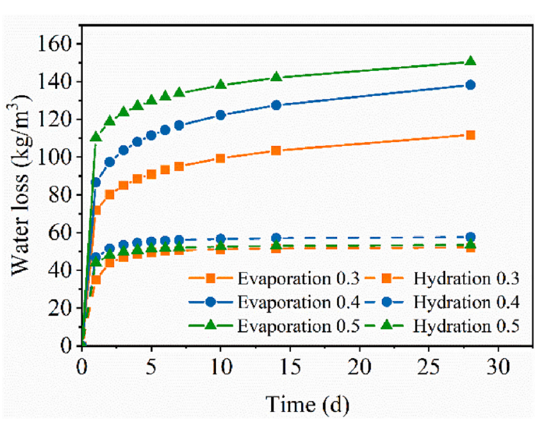


Fig. 20. Comparison between evaporated and hydrated water of the printed specimens with varying w/c.

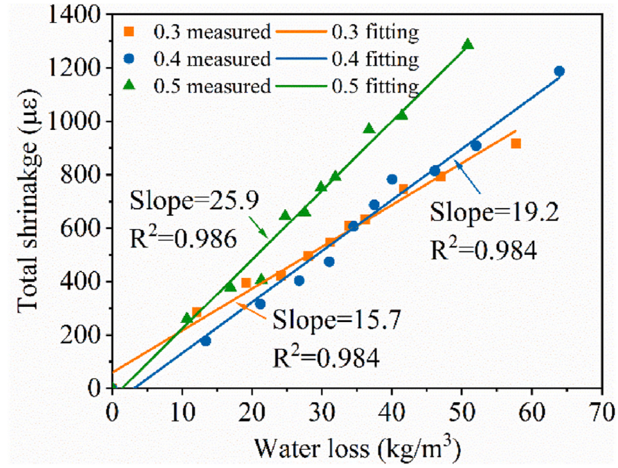


Fig. 21. Correlation between water loss (evaporation and hydration) and total shrinkage.

5.3. Total shrinkage modeling of 3DPC

In the preceding sections, we have identified from the experimental perspective the impact of varying w/c on the shrinkage of 3DPC under dry conditions. Besides these results, there is still a notable gap in the modeling research concerning the shrinkage of 3DPC in such environments. This section aims to develop a shrinkage model for 3DPC. Although the underlying mechanisms of shrinkage in 3DPC are similar to those in cast concrete, the unique manufacturing process of 3DPC poses specific characteristics in establishing an accurate shrinkage model. Therefore, existing models for cast concrete were refined to better align with the performance characteristics of 3DPC, particularly considering its direct exposure to dry environment where hydration and evaporation occur simultaneously. The process of establishing the model began with the introduction of total shrinkage modeling in Section 2. Subsequently, the detailed calculation of major parameters, including the saturation degree (S), capillary pressure (P), and bulk modulus (K_b), was carried out. Finally, a comparison of the predicted results with actual measurements was presented.

The whole prediction process of total shrinkage is illustrated in Fig. 22. In the diagram, the parameters and the involved equations are also shown in each parameter block. The model parameters determined by experiments in this study are saturation degree (S), capillary pressure (P), and bulk modulus (K_b). In the case of 3DPC, early exposure to drying conditions results in concurrent processes of hydration and evaporation. Therefore, it is necessary to simultaneously account for the effects of both hydration and evaporation on the saturation degree. To calculate the saturation degree, we quantified the free water content based on the principle of water mass conservation within the concrete. Therefore, the saturation degree was determined through calculations involving mass loss (evaporated water), hydration degree (hydrated water), and porosity. Capillary pressure was influenced by internal RH and could be calculated using the Kelvin-Laplace equation. The bulk modulus was primarily related to the elastic modulus of the specimen and could be derived by fitting the hydration degree. The specific calculation process for these parameters will be elaborated in subsequent sections.

5.3.1. Capillary pressure

The generation of capillary pressure in 3DPC is primarily associated with water loss. This consumption disrupts the continuity of moisture within the concrete, causing moisture to migrate into the pores, and negative pressure is generated, known as capillary pressure [37]. The relationship between capillary pressure and the internal RH can be described using the Kelvin-Laplace equation [67] :

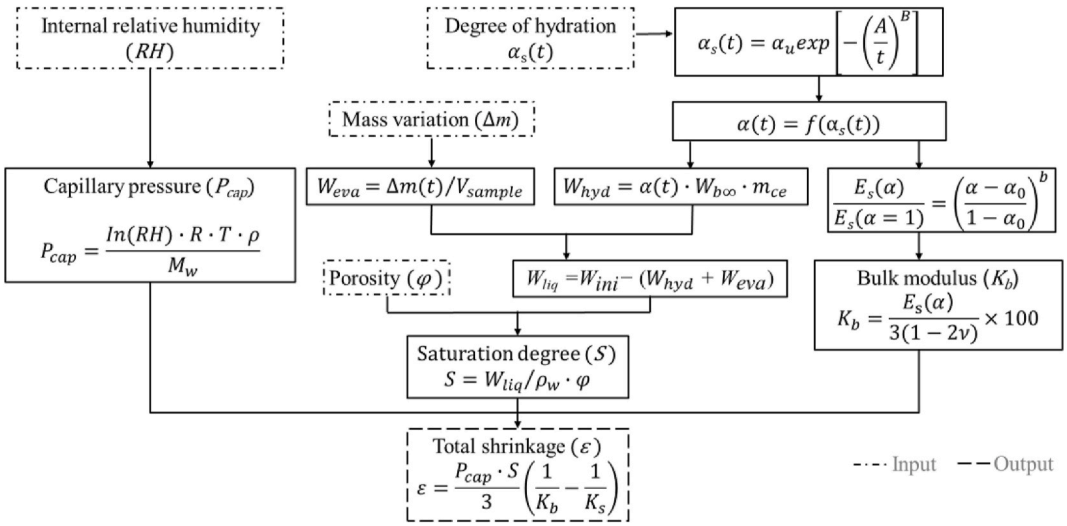


Fig. 22. Roadmap for shrinkage model establishment of 3DPC.

$$P_{cap} = \frac{In(RH) \bullet R \bullet T \bullet \rho}{M_w} \tag{20}$$

where, R is the universal gas constant, 8.314 J/(mol·K); T is the absolute temperature, 293.15 K; M_w is the molar mass of water, 0.01802 kg/mol; ρ is the density of water, 1000 kg/m^3. Results for RH can be obtained in section 4.5.

5.3.2. Hydration degree

The degree of hydration, essential for calculating the saturation degree and bulk modulus, is determined under standard curing conditions using isothermal calorimetry testing. The specifics of the testing and calculation process are elaborated in section 3.5.2. The major parameters of the hydration degree in the standard condition can be presented in Table 9.

The development trend of the hydration degree under standard conditions can be further fitted using Eq. (21) [68]:

$$\alpha = \alpha_u \, exp\left[-\left(\frac{A}{t}\right)^B\right] \tag{21}$$

where, α_u is the final hydration degree of cement; t is the equivalent age of concrete at 20 °C. Since the temperature in this study is 20 °C, the actual age is the equivalent age; A and B are fitting parameters. The final hydration degree (α_u) depends on the w/c and can be calculated using Eq. (22):

$$a_u = \frac{\frac{1.031w}{b}}{0.194 + \frac{w}{b}} \tag{22}$$

The degree of hydration under standard curing conditions is calculated using Eq. (21), with the key parameters detailed in Table 10. To determine the hydration degree under drying conditions, thermal gravimetric testing is utilized, as described in section 4.7. Consequently, this enables us to determine the correlation between the degrees of hydration under drying conditions and standard curing conditions, which is graphically represented in Fig. 23. By establishing the relationship between the degree of hydration under dry and standard conditions, we can track the progression of hydration over time in a dry environment.

5.3.3. Saturation degree

The degree of saturation is determined as the ratio of the evaporable water content (V_{ew}) to the total pore volume of the concrete (V_p) [69]. There are three methods to test the degree of saturation. The first method is the mass change method, this method involves calculating the degree of saturation by measuring the mass loss of the specimen at time 't' compared to its initial mass ($\Delta m(t)$) and calculating the degree of saturation based on the porosity (φ) and water density (ρ_w). The formula is given by Eq. (23) [70]:

$$S = 1 - \frac{\Delta m(t)}{\varphi v \rho_w} \tag{23}$$

The second method measures the mass of the specimen in different environmental conditions (initial, dry, and saturated) and calculates the degree of saturation. The formula is given by Eq. (24) [71]:

$$S = \frac{V_{ew}}{V_p} = \frac{m_i - m_d}{m_s - m_d} \tag{24}$$

where, m_i is the initial mass of the sample; m_d is the mass of the sample at dry environment; m_s is the mass of the sample at saturated environment.

The third method is the Powers model. The Powers model is based on the empirical model proposed by Powers and Brownyard in 1948 [72]. It calculates the degree of saturation based on the composition and distribution of phases in hardened cement paste. This model describes a function of w/c and hydration degree (α), which can be given by Eq. (25):

$$S = \frac{V_{ew}(\alpha)}{V_p(\alpha)} = \frac{V_{cw}(\alpha) + V_{gw}(\alpha)}{V_{cw}(\alpha) + V_{gw}(\alpha) + V_{cs}(\alpha)} \tag{25}$$

Table 9
Calculated parameters of the degree of hydration in standard condition.

w/c	C (kJ/kg °C)	ΔT (∞) (°C)	H_0 (kJ/kg)
0.3	1.71	216.16	480.49
0.4	1.87	180.45	480.49
0.5	1.88	144.36	480.49

Table 10
Fitting parameters on the development of the degree of hydration.

w/c	α_u	A	B	R^2
0.3	0.626	0.755	1.289	0.999
0.4	0.694	0.669	1.002	0.997
0.5	0.743	0.617	0.955	0.998

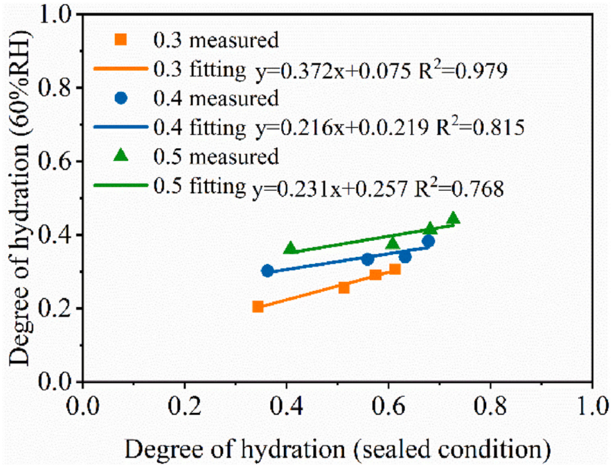


Fig. 23. Correlation between the hydration degree under 60% RH and that in standard condition.

where, V_{cw}, V_{gw}, and V_{cs} are functions of α and represent different water content contributions from capillary pore water volume, gel water volume, and chemical shrinkage, respectively.

Mass change methods are particularly applicable for concrete that has been exposed to long-term dry conditions, wherein the saturation degree is primarily influenced by evaporation, and the impact of hydration is comparatively less pronounced. The Powers model is employed to evaluate the saturation of concrete specimens in sealed conditions, where the saturation is mainly governed by the hydration process. In the case of 3DPC, which is immediately exposed to dry environment, it becomes crucial to account for both hydration and evaporation effects. Therefore, our approach involves calculating the saturation degree by integrating considerations of both hydration and evaporation processes. The water mass conservation principle includes four water types in concrete: hydrated water (W_{hyd}), evaporable water (W_{eva}), liquid water in pores (W_{liq}), and water vapor in pores (W_{vap}). Given the relatively low content of water vapor in the pore structure, its impact on the saturation can be considered negligible. Following the principle of water mass conservation, the internal water content of the concrete can be represented by Eq. (26):

$$W_{ini} = W_{hyd} + W_{eva} + W_{liq} \tag{26}$$

To calculate the degree of saturation for 3DPC, both hydration-induced water loss and evaporation-induced water loss need to be considered. The relationship between free water content, hydration degree, and mass loss can be established using Eqs. (18) and (19). Substituting these Equations into Eq. (26) allows for the calculation of free water content as a difference between initial water content and the sum of hydration- and evaporation-induced water losses (Eq. (27)) [73].

$$W_{liq} = W_{ini} - \left(W_{hyd} + W_{eva}\right) = W_{ini} - \left(\alpha(t) \bullet W_b \bullet m_{ce} + \frac{\Delta m}{V_{sample}}\right) \tag{27}$$

In this way, the relationship between the free water content, mass loss and degree of hydration can be established. The main parameter, initial water content (W_{ini}), is detailed in Table 4. With the results of porosity (φ), outlined in section 4.8, the degree of saturation can be calculated using Eq. (28):

$$S = \frac{\frac{w_{liq}}{\rho_w}}{\varphi} \tag{28}$$

5.3.4. Bulk modulus

The volume modulus (K) of concrete can be determined using Young's modulus (E) and Poisson's ratio (ν) and is expressed in Eq. (29):

$$K = \frac{E_s}{3(1-2\nu)} \tag{29}$$

where, K is the volume modulus of the specimen, GPa; E_s is the Young's modulus, GPa; ν is the Poisson's ratio, 0.2 [74]. To determine the development of the static elastic modulus over time, its relationship with the degree of hydration is further established according to Eq. (30) [75]:

$$\frac{E(\alpha)}{E(\alpha=1)} = \left(\frac{\alpha-\alpha_0}{1-\alpha_0}\right)^b \tag{30}$$

where, $E(\alpha)$ is the elastic modulus at hydration degree of α; $E(\alpha=1)$ is the elastic modulus of the specimen when fully hydrated; α_0 and b are the fitting parameters derived from the relationship between elastic modulus and hydration degree. The outcomes of the fitting process are depicted in Fig. 24, while the parameters derived from this fitting are detailed in Table 11.

5.3.5. Comparison of measured and modeled shrinkage

Based on previous calculations of capillary pressure, saturation, and bulk modulus, the shrinkage prediction of 3DPC can be established using the Biot-Bishop model:

$$\varepsilon = \frac{P_{cap} \bullet S}{3}\left(\frac{1}{K_b} - \frac{1}{K_s}\right) \tag{31}$$

where, P_{cap} is capillary pressure, GPa; S is saturation degree; K_b is bulk modulus of the porous material, GPa; K_S is bulk modulus of the solid body, in the context of cement paste, K_S typically ranges from 38 to 55 GPa. This variance largely depends on the bulk modulus of all the solid phases within the system. The most significant uncertainty arises from the bulk modulus of C–S–H due to its inherent porosity [76]. Given that the formation process of 3DPC does not alter the bulk modulus of C–S–H, this study adopts the mean value commonly used in most research. Therefore, K_S for the fully solid body is considered to be 44 GPa.

The comparison between the modeling results using Eq. (31) and actual measured values is shown in Fig. 25. The predicted shrinkage of 3DPC generally aligns with the actual measurements, particularly in the initial stages for specimens with varying w/c of 0.3, 0.4, and 0.5. However, discrepancies arise after 10 d, with the model tending to underestimate shrinkage compared to the measured values. This divergence may be rooted in the assumptions and simplifications inherent in the modeling process. For instance, the model might not fully capture complex interactions between hydration and evaporation processes in 3DPC, or it may oversimplify the concrete response to environmental factors over time. The assumptions made in modeling capillary pressure and its relation to internal RH, or in the calculation of the saturation

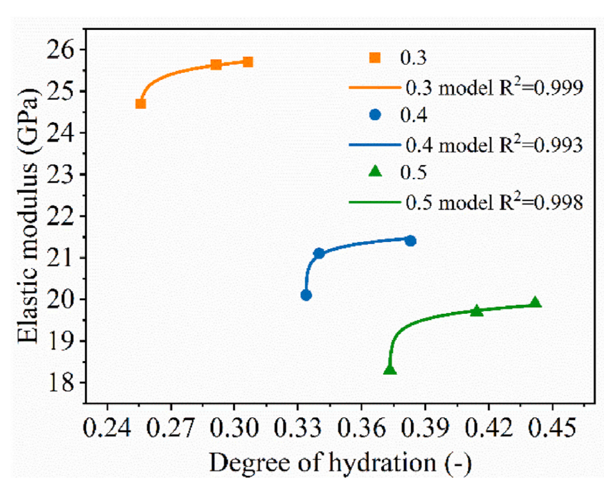


Fig. 24. Elastic modulus determined by fitting to the degree of hydration.

Table 11
Main parameters of elastic modulus fitting based on the degree of hydration.

w/c	E ($\alpha = 1$)	α_0	b	R^2
0.3	26.4	0.25	0.02	0.999
0.4	22.0	0.33	0.02	0.993
0.5	20.4	0.37	0.03	0.998

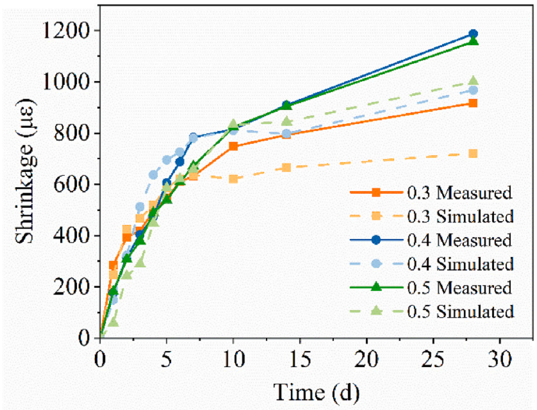


Fig. 25. Comparison of simulated and tested total shrinkage of 3DPC.

degree, might also contribute to this deviation.

While the overall error margin remains within 20%, this figure is significant given that the measured shrinkage difference between samples with w/c from 0.3 to 0.4 and from 0.4 to 0.5 is also around 20%. This similarity in error margin suggests that the model, while effective in general terms, requires refinement to enhance its predictive accuracy, particularly for long-term behavior.

Future improvements could focus on more accurately modeling the interplay of hydration and evaporation, refining calculations of capillary pressure, elucidating the role of autogenous shrinkage and better accounting for the dynamic environmental conditions affecting 3DPC. In conclusion, the model demonstrates a promising capability to predict 3DPC shrinkage, yet highlights spaces for refinement, especially in its long-term predictive accuracy and the need to account for the nuanced behaviors of 3DPC under varying environmental conditions.

6. Conclusion

This study explored how different initial moisture conditions (w/c) and the exposed surface area (s/v) affect water loss, pore structure, and shrinkage in 3DPC. The satisfactory printability of concretes with different w/c was achieved by incorporating calcium formate into high w/c mixtures in order to make reasonable and meaningful comparison of the results.

Of all the parameters investigated in this study (w/c, s/v, and cast vs. 3D printing), it was observed that w/c has the most significant impact on total shrinkage. The total shrinkage of printed specimens increases by 37.3% with the w/c increasing from 0.3 to 0.5. In contrast, the rise of s/v from 0.06 to 0.17 increases the total shrinkage of printed specimens by 19.8%. The total shrinkage of 3DPC specimens with low w/c (0.3) is comparable to that of their cast counterparts, exhibiting only a marginal 4.7% discrepancy. However, when the w/c increases from 0.3 to 0.5, the difference in shrinkage between printed and cast specimens increases to 10%. This increased difference in shrinkage could be explained by the higher pore volume in specimens with higher w/c, making them more susceptible to the effects of vibration and densification processes during casting.

Furthermore, we conducted a quantitative analysis to compare evaporated and hydrated water in 3DPC, focusing on how early-stage drying affects water loss. The water consumed by evaporation exceeds that consumed by hydration, with the water evaporated at 28 d up to 2.8 times higher than the water consumed by hydration. By evaluating the total shrinkage of specimens with different w/c and conducting a comprehensive analysis of internal RH, mass loss, degree of hydration, and porosity, a model to predict the total shrinkage is developed for 3DPC. Comparisons with experimental data demonstrated that the predictions align closely with actual results, maintaining an error margin of 20%.

CRediT authorship contribution statement

Lei Ma: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Zijian Jia:** Methodology. **Yuning Chen:** Methodology. **Yifan Jiang:** Methodology. **Bruno Huet:** Methodology. **Arnaud Delaplace:** Methodology. **Yamei Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Qing Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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